

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech.

Branch: Automobile/ Production/Mechanical Engineering.

Semester : VIII

Subject Code: BTMEC801F/ BTAMC801F

Subject Name: Non-Conventional Energy Resources

Max Marks: 60

Date: 07/07/2022

Duration: 3.45 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- | | |
|---|---|
| A) What is fossil fuel? What are different alternatives for fossil fuel? | 6 |
| B) Explain national energy strategies and National energy plan | 6 |
| C) Explain energy consumption as a measure of prosperity and world energy future. | 6 |

Q.2 Solve Any Two of the following.

- | | |
|---|---|
| A) Explain solar energy as alternative energy source. | 6 |
| B) Explain solar energy conversion systems and their applications. | 6 |
| C) Explain with neat sketch solar flat plate collector as solar air heater. | 6 |

Q. 3 Solve Any Two of the following.

- | | |
|--|---|
| A) Explain the following terms
1. Solar constant.
2. Solar spectrum.
3. Clarity Index.
4. Declination angle.
5. Zenith angle.
6. Day length hours. | 6 |
| B) Explain principle of working of a solar cell. | 6 |
| C) Explain various types of commercial solar cells. | 6 |

Q.4 Solve Any Two of the following.

- | | |
|--|---|
| A) What is wind data and energy estimation in wind energy? | 6 |
| B) What are various types of rotors in wind mill? Draw a neat labelled sketch of propeller type of wind machine. | 6 |
| C) What is principle of OTEC? Draw neat labelled sketch of open cycle OTEC power plant. | 6 |

Q. 5 Solve Any Two of the following.

- | | |
|---|---|
| A) Explain principle of MHD power generation. | 6 |
| B) Write design and principle of operation of a fuel cell. | 6 |
| C) Explain with neat sketch working of Lithium ion Batteries. | 6 |

***** End *****